

Principles of Communication Systems

Mid-Term Examination

Instructor: Ranjitha Prasad

Duration: 60 minutes Total: 20 marks

Please provide your answers clearly, with all the steps outlined. Some of the trigonometric identities required are as follows:

- $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$
- $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$
- $\cos(\theta_1 - \theta_2) = \cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)$

Q1: 6 marks, Q2: 7 marks, Q3: 4 marks, Q4: 3 marks

Problem 1 Assume that $m(t)$ is a real baseband signal with bandwidth W , and

$$x_p(t) = m(t)\cos(2\pi f_0 t) \quad y_p(t) = m(t)\sin(2\pi f_0 t),$$

where $f_0 \gg W$. Compute the inner-products in the passband and the complex baseband, i.e., compute $\langle x_p, y_p \rangle$ and $\langle x, y \rangle$, where $x(t)$ and $y(t)$ is the complex baseband representation of $x_p(t)$ and $y_p(t)$, respectively. Can you comment about orthogonality in passband and orthogonality in complex baseband. ... (6)

Problem 2 Consider the transmitter as given below in Fig. a. Let the demodulator be of the form in Fig. b. Provide expression for A and plot $H(f)$ ($H(f) \leftrightarrow h(t)$) if ... (7)

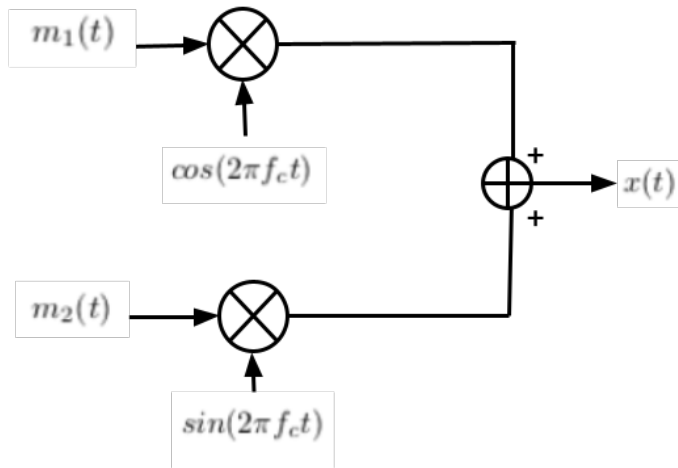


Fig. a

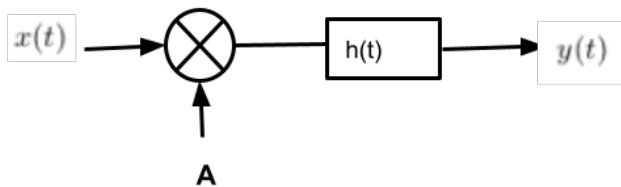


Fig. b

Figura 1: Fig.a and Fig. b for Question 2

- We need the output $y(t) = m_1(t)$.
- We need the output $y(t) = m_2(t)$.

Problem 3 In the receiver shown below, local oscillator operates at a frequency (f_{LO}) below the incoming carrier frequency f_c . Given $f_{IF} = 30MHz$. .. (4)

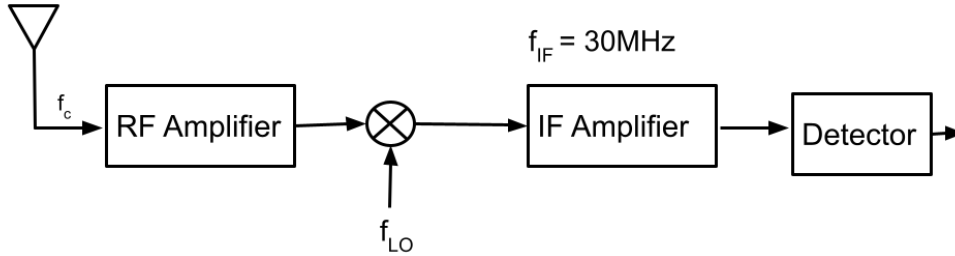


Figura 2: Question 3

- If the receiver is designed to receive signals from 120 – 150MHz, what is the tuning range required for the local oscillator?
- What is f_{LO} when a 136MHz carrier signal is received. What is the image frequency for this carrier frequency.

Problem 4 Consider the FM signal with $k_f = 20\pi$. The message signal $m_1(t)$ is as given below. Plot the frequency deviation of the FM signal as a function of time. ... (3)

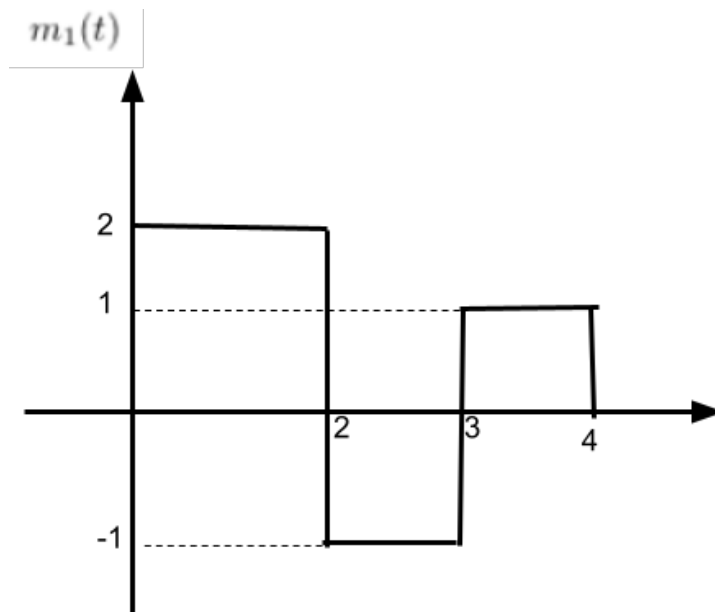


Figura 3: Question 4